

2024



Jonas Fitness Sunsets Mainframe Using AWS Mainframe Modernization Automated Refactor

Learn how software company Jonas Fitness refactored its legacy mainframe applications for improved performance on AWS.

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Reduced costs

by 90%

Achieved

within 1 month



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Adopted

modern development practices



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For over 30 years, [Jonas Fitness Inc.](#) (Jonas Fitness) has offered integrated payment systems to help gyms and other wellness centers manage their membership programs. Originally coded by Jonas Fitness's founder, the company's software encompasses several applications that were running on an aging legacy mainframe, which stifled its potential for growth.

Recognizing the need to modernize, Jonas Fitness sought a solution that would upgrade its infrastructure while preserving the functionality of its software. Turning to Amazon Web Services (AWS), the company completely refactored its mainframe applications on the cloud, improving efficiency and agility.



Opportunity | Using AWS Blu Age to Modernize Jonas Fitness's Mainframe-Based Payment Applications

Founded in 1981, Jonas Fitness provides club management software to over 3,000 gyms, health clubs, and wellness centers. As one of the first companies to offer this solution, its founder had to build everything from scratch. Consequently, Jonas Fitness has long relied on a legacy mainframe programmed using Common Business Oriented Language (COBOL).

"The legacy mainframe is the core of our business, but it has significantly aged. It was getting harder to find the right talent to help build and maintain our applications," says Tony Autin, president of Jonas Fitness. "In addition, our legacy hosting provider was expensive."

Faced with these mounting challenges, Jonas Fitness initiated a search for a more cost-effective, modern solution. The company explored offerings from various vendors, ultimately discovering that [Blu Age](#), a software vendor that specializes in sunseting mainframe technologies, resonated best with its needs. Blu Age was later acquired by AWS.

"We engaged AWS Blu Age in a proof of concept that focused on refactoring one mainframe application: our payment gateway," says Autin. "That project was successful, so we initiated the full modernization of our mainframe."



Solution | Reducing Costs by 90% with Refactored Applications on AWS

From April 2022 to May 2023, Jonas Fitness modernized its mainframe using [AWS Mainframe Modernization Automated Refactor with AWS Blu Age](#). AWS Blu Age powers the automated refactoring pattern within the [AWS Mainframe Modernization](#) service, which customers use to migrate, modernize, run, test, and operate mainframe workloads. This service offers prepackaged, fully managed cloud-native environments with preintegrated toolchains. AWS Mainframe Modernization Automated Refactor accelerated the modernization of the applications' complete software stack, including application code, dependencies, and infrastructure.

During the refactoring process, data was migrated from the company's legacy data center to the cloud. Business logic coded in COBOL was converted to Java, and IBM Db2 application data and schemas were migrated to a PostgreSQL database with high precision and performance. Additionally, interactive legacy IBM customer information control systems screens were transformed into powerful single-page Angular web applications.

Jonas Fitness's applications were then completely refactored using equivalent AWS services. For example, security authentication and authorization are now managed by [Amazon Cognito](#), a solution used to implement secure, frictionless customer identity and access management at scale. The company also adopted [Amazon Aurora for PostgreSQL](#), a relational database management system, to support its primary PostgreSQL database. "Using an open-source database saves costs, but we also use Amazon Aurora for PostgreSQL to gain redundancy, improve performance, and simplify management," says James Peltier, CTO of Jonas Fitness. "We don't have to manage underlying servers, which is a major advantage from a maintenance standpoint."

Over a single weekend in May 2023, Jonas Fitness officially transitioned from the mainframe to AWS. While preparing for the changeover, however, the company had to run two environments at the same time. To manage the costs associated with this setup, Jonas Fitness participated in the [AWS Migration Acceleration Program](#) (AWS MAP), a program that helps accelerate cloud migration and modernization journeys with an outcome-driven methodology.

"Running two environments was a very expensive proposition for us, especially as we prepared to go live," says Peltier. "AWS MAP helped us mitigate these up-front costs. Without these types of programs, achieving a return on investment is much more difficult."

Jonas Fitness has reaped significant technical benefits from the transition. By using Java instead of COBOL, the company can now adopt modern software development practices. For example, it now runs applications in Docker containers managed by [AWS Fargate](#), a serverless, pay-as-you-go compute engine. This shift has increased agility and performance; Jonas Fitness can decouple applications from the main architecture and scale as needed. Now, the company can optimize its system at the application level, whether it be for security, business, or efficiency.



The cloud architecture has also reduced Jonas Fitness’s technical debt. The company no longer needs to run the entire mainframe 24 hours per day, 7 days per week, nor does it need to spend time on management or maintenance. Instead, it can scale up or down based on demand. With this combination of benefits, Jonas Fitness has reduced costs by 90 percent. “Running our mainframe on AWS is 10 times less expensive than what it used to cost,” says Autin. “We typically target a 30 percent return on investment, but this project yielded over 100 percent in less than 1 year—which is massively significant for our business.”

Outcome | Exploring Advanced Technologies to Better Serve Customers

Looking ahead, Jonas Fitness aims to take further advantage of its modern, scalable architecture. With plans to integrate cutting-edge technologies such as data lakes and artificial intelligence, the company’s focus remains firmly on enhancing the customer experience. “We really want to help our customers and find ways to make their lives a little easier,” says Autin.

By refactoring its mainframe using AWS Blu Age, Jonas Fitness has modernized its infrastructure and established a solid foundation for future growth. The migration has energized the company’s workforce by equipping them with modern technologies, which they are using to explore new applications. This has not only helped Jonas Fitness improve the experience of its current employees, but has also facilitated the recruitment of new talent. With the agility of the cloud, the company is poised to explore new capabilities and enhance value for its customers.

About Jonas Fitness Inc.

Jonas Fitness Inc. (Jonas Fitness) has been a provider of club management software used by gyms, health clubs, and wellness centers for more than 30 years. Serving over 3,000 locations, Jonas Fitness processes more than 1.4 billion credit card transactions per year.

AWS Services Used

AWS Mainframe Modernization

AWS Mainframe Modernization service is a unique platform that allows you to migrate and modernize your on-premises mainframe applications to a cloud native fully-managed runtime environment on AWS.

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Amazon Cognito



Amazon Cognito provides an identity store that scales to millions of users, supports social and enterprise identity federation, and offers advanced security features to protect your consumers and business.

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Amazon Aurora for PostgreSQL

Amazon Aurora is a relational database service that combines the speed and availability of high-end commercial databases with the simplicity and cost-effectiveness of open-source databases. Aurora is fully compatible with MySQL and PostgreSQL, allowing existing applications and tools to run without requiring modification.

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AWS Fargate

AWS Fargate is a serverless, pay-as-you-go compute engine that lets you focus on building applications without managing servers.

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